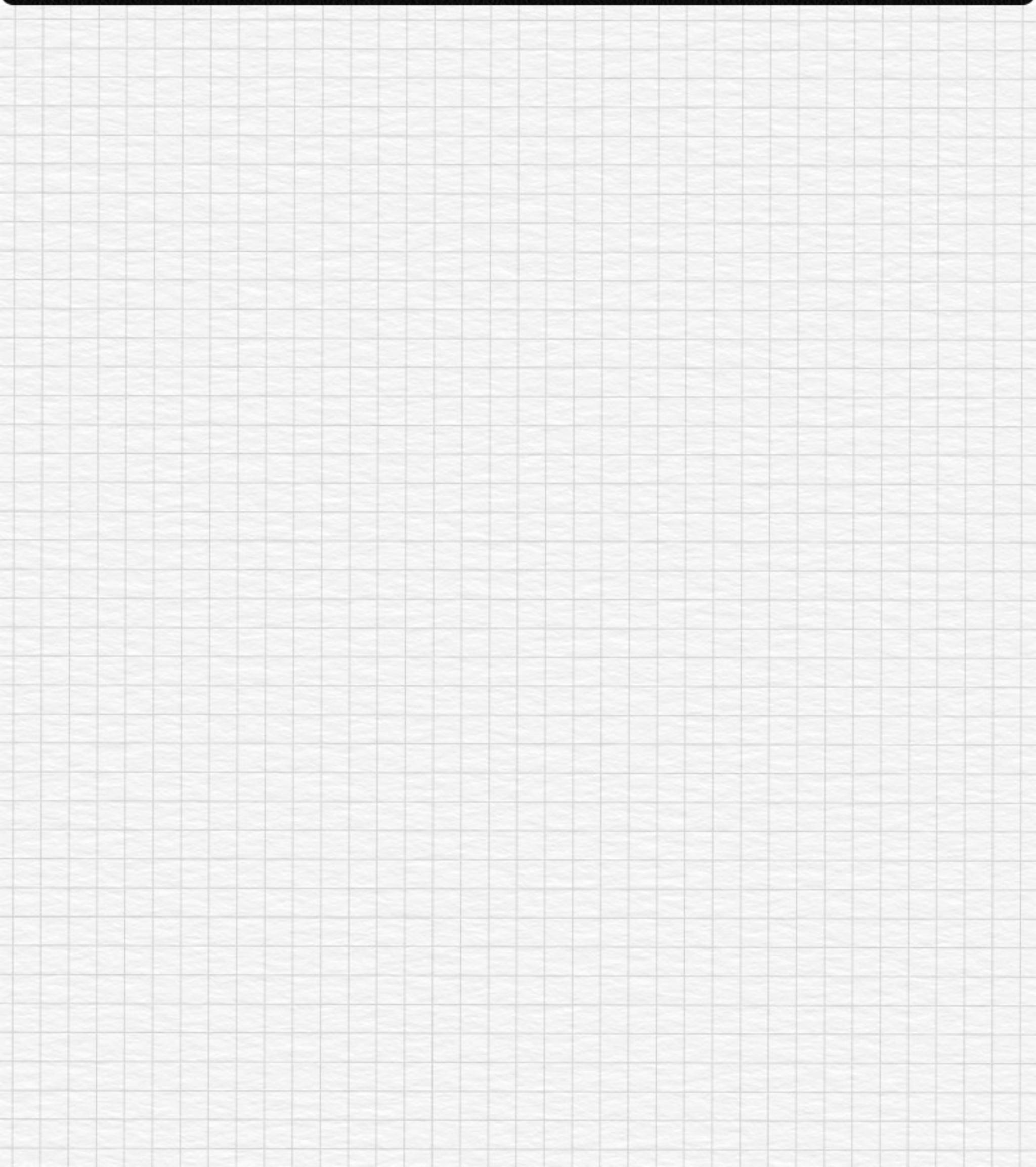
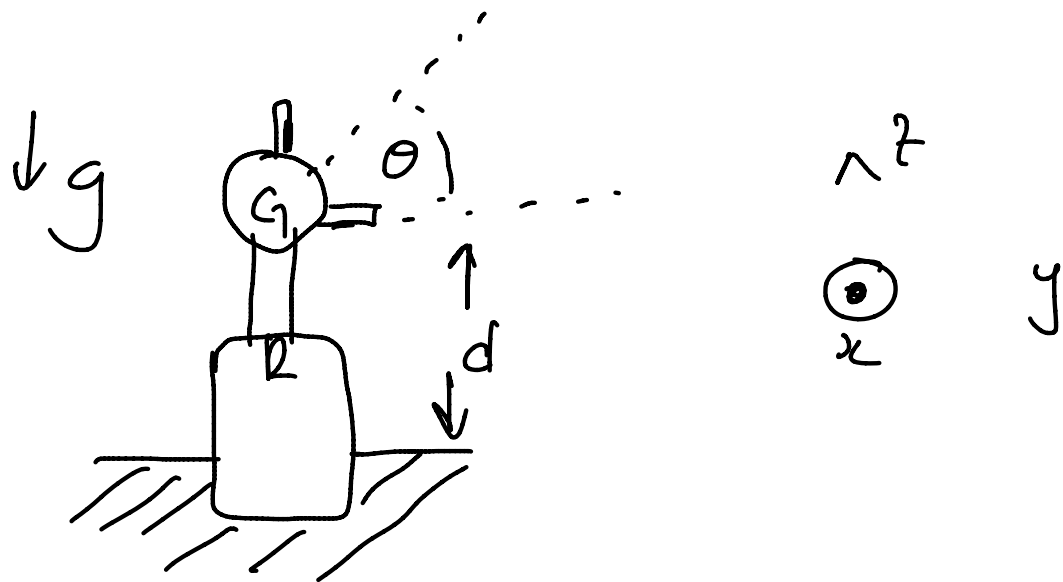


NOTEBOOK



Problem 1



m_g — mass of gripper

m_r — mass of rod

I — moment of inertia

Kinetic Energy

$$T = \frac{1}{2} I \dot{\theta}^2 + \frac{1}{2} m_g \dot{d}^2 + \frac{1}{2} m_r \left(\frac{\dot{d}}{2} \right)^2$$

Potential Energy

$$U = m_g g d + m_r g \frac{d}{2}$$

$$\therefore T = \frac{1}{2} I \dot{\theta}^2 + \frac{1}{2} \left(m_g + \frac{1}{4} m_r \right) \dot{d}^2$$

$$U = \left(m_g + \frac{1}{2} m_r \right) g d$$

The Lagrangian, $L = T - U$

$$\therefore L = \frac{1}{2} I \dot{\theta}^2 + \frac{1}{2} (m_G + \frac{1}{4} m_R) \dot{d}^2 - (m_G + \frac{1}{2} m_R) g d$$

Euler-Lagrange

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$$

$$\frac{\partial L}{\partial \theta} = 0$$

$$\frac{\partial L}{\partial d} = - (m_G + \frac{1}{2} m_R) g$$

$$\frac{\partial}{\partial t} \left(\frac{\partial L}{\partial \dot{\theta}} \right) = I \ddot{\theta}$$

$$\frac{\partial}{\partial t} \left(\frac{\partial L}{\partial \dot{d}} \right) = (m_G + \frac{1}{4} m_R) \ddot{d}$$

Two equations

$$I \ddot{\theta} = \tau_1$$

$$(m_G + \frac{1}{4} m_R) \ddot{d} - (m_G + \frac{1}{2} m_R) g = f_2$$

$$\therefore \begin{bmatrix} I & 0 \\ 0 & m_G + \frac{1}{4}m_R \end{bmatrix} \cdot \begin{bmatrix} \ddot{\theta} \\ \ddot{d} \end{bmatrix} + \begin{bmatrix} 0 \\ -(m_G + \frac{1}{2}m_R)g \end{bmatrix} = \begin{bmatrix} \tau \\ f \end{bmatrix}$$

$$B(q) \cdot \ddot{q} + G(q) = \tau$$

There are no :-

$C(q) \Rightarrow$ no relative rotation

$F_{S/u} \Rightarrow$ no friction

$J^T(q) \Rightarrow$ no interaction w/ environment

Euler-Lagrange Equations

$$I \ddot{\theta} = \tau$$

$$(m_G + \frac{1}{4} m_R) \ddot{d} - (m_G + \frac{1}{2} m_R) g = \vec{f}_2$$

$$\begin{bmatrix} \ddot{\theta} & 0 & 0 \\ 0 & \ddot{d} & g \end{bmatrix} \begin{bmatrix} I \\ m_G + \frac{1}{4} m_R \\ m_G + \frac{1}{2} m_R \end{bmatrix} = \begin{bmatrix} \tau_1 \\ \vec{f}_2 \end{bmatrix}$$

$$Y \cdot \tau = \dot{c}$$

$$Y = \begin{bmatrix} \ddot{\theta} & 0 & 0 \\ 0 & \ddot{d} & g \\ 0 & g & \ddot{d} \end{bmatrix} \quad Y^T = \begin{bmatrix} \ddot{\theta} & 0 & 0 \\ 0 & \ddot{d} & g \\ 0 & g & \ddot{d} \end{bmatrix}$$

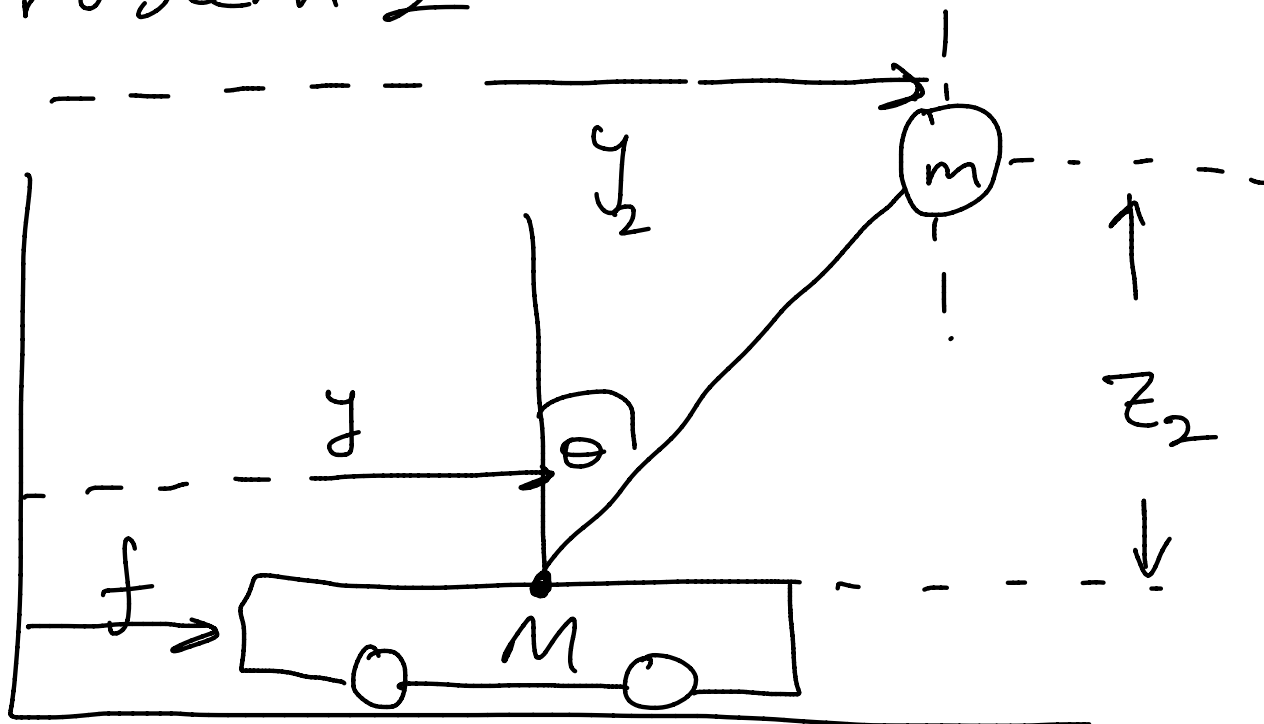
$$Y^T \cdot Y = \begin{bmatrix} \ddot{\theta}^2 & 0 & 0 \\ 0 & \ddot{d}^2 & g\ddot{d} \\ 0 & g\ddot{d} & g^2 \end{bmatrix}$$

$$\det(Y^T Y) = \ddot{\theta}^2 (g^2 \ddot{d}^2 - g^2 \ddot{d}^2) = 0$$

Since the determinant is zero, the matrix cannot be inverted.

Therefore individual values of π cannot be approximated.

Problem 2



M - mass of cart

l - length of pendulum

m - mass of bob

Position of Bob

$$\mathbf{r} = \begin{bmatrix} 0 \\ l \sin \theta \\ l \cos \theta \end{bmatrix}$$

Velocity of Bob (Rotational)

$$\dot{\mathbf{r}} = \begin{bmatrix} 0 \\ \dot{\theta} l \cos \theta \\ -\dot{\theta} l \sin \theta \end{bmatrix}$$

$$\begin{aligned} \|\dot{\mathbf{r}}\|^2 &= (\dot{\theta} l \cos \theta)^2 + (-\dot{\theta} l \sin \theta)^2 \\ &= \dot{\theta}^2 l^2 \end{aligned}$$

Kinetic Energy

$$\begin{aligned} T &= \frac{1}{2} m \dot{\theta}^2 l^2 + \frac{1}{2} m \dot{y}^2 + \frac{1}{2} M \dot{y}^2 \\ &= \frac{1}{2} m \dot{\theta}^2 l^2 + \frac{1}{2} (m+M) \dot{y}^2 \end{aligned}$$

Potential Energy

$$U = mgl \cos \theta$$

Lagrange

$$L = T - U = \frac{1}{2} m \dot{\theta}^2 l^2 + \frac{1}{2} (m+M) \dot{y}^2 - mgl \cos \theta$$

Euler - Lagrange

$$m l^2 \ddot{\theta} - mgl \sin \theta = \tau$$

$$(m+M) \ddot{y} = f$$

Dynamics Equations

$$\begin{bmatrix} m l^2 & 0 \\ 0 & m+M \end{bmatrix} \begin{bmatrix} \ddot{\theta} \\ \ddot{y} \end{bmatrix} + \begin{bmatrix} -mgl \sin \theta \\ 0 \end{bmatrix} = \begin{bmatrix} \tau \\ f \end{bmatrix}$$

$$B(q) \cdot \ddot{q} + G(q) = \tau$$

" $\gamma - \pi$ " form

$$\begin{bmatrix} l^2 \ddot{\theta} - gl \sin \theta & 0 \\ 0 & \ddot{y} \end{bmatrix} \begin{bmatrix} m \\ m+M \end{bmatrix} = \begin{bmatrix} \tau \\ f \end{bmatrix}$$

$\gamma \qquad \qquad \pi \qquad \qquad \tau$

γ is symmetric, so $\gamma^T = \gamma$

$$\gamma^T \gamma = \begin{bmatrix} (l^2 \ddot{\theta} - gl \sin \theta)^2 & 0 \\ 0 & \ddot{y}^2 \end{bmatrix}$$

determinant = $(l^2 \ddot{\theta} - gl \sin \theta)^2 - \ddot{y}^2$

invertible $\forall \ddot{\theta}, \theta, y$ where determinant $\neq 0$

$$(\gamma^T \gamma)^{-1} = \frac{1}{(l^2 \ddot{\theta} - gl \sin \theta)^2 - \ddot{y}^2} \begin{bmatrix} \ddot{y}^2 & 0 \\ 0 & (l^2 \ddot{\theta} - gl \sin \theta)^2 \end{bmatrix}$$

$\therefore \pi$ can be approximated as

$$\hat{\pi} = (\gamma^T \gamma)^{-1} \gamma^T \cdot \tau$$